

LBEADS-NET

Algorithm Unrolling for Learned Baseline Estimation
and Denoising in Chromatographic Signals

MS Thesis Defense

Saad Zubairi

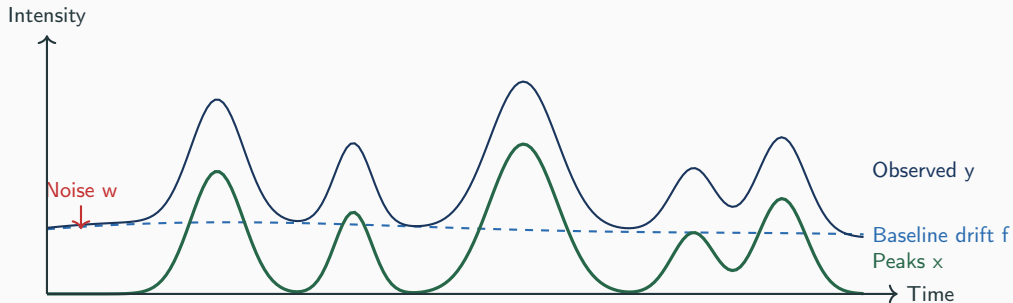
Advisor: Professor Ivan Selesnick

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New York University
Tandon School of Engineering

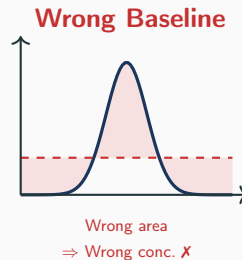
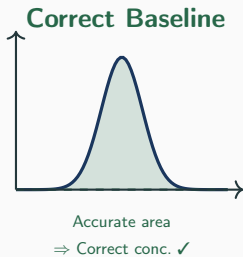
Act I: The Problem

What is a Chromatogram?



The observed signal is a mixture of three components: $y = x + f + w$

Why Baseline Correction Matters



In pharmaceutical QC, even 2% baseline error can exceed regulatory limits.

The Classical Solution: BEADS

Baseline **E**stimation **A**nd **D**enoising using **S**parsity

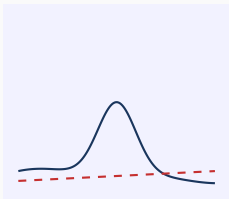
(Ning et al., 2014)

$$J(x) = \frac{1}{2} \|H(y - x)\|_2^2 + \lambda_0 \sum_n \theta(x_n) + \lambda_1 \sum_n \phi(\Delta x_n) + \lambda_2 \sum_n \phi(\Delta^2 x_n)$$

1. **Data fidelity** — high-pass filter H separates baseline from peaks
2. **Asymmetric sparsity penalty** θ — peaks are sparse and non-negative
3. **First-order smoothness** — penalizes peak roughness
4. **Second-order smoothness** — penalizes peak curvature

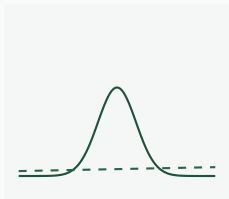
6 parameters to tune: $\lambda_0, \lambda_1, \lambda_2, r, f_c, d$

The Tuning Problem



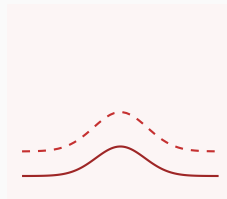
Under-regularized

Noisy, baseline too low



Well-tuned

Clean separation



Over-regularized

Peaks absorbed into baseline

No single parameter configuration generalizes across instruments, analytes, or conditions.

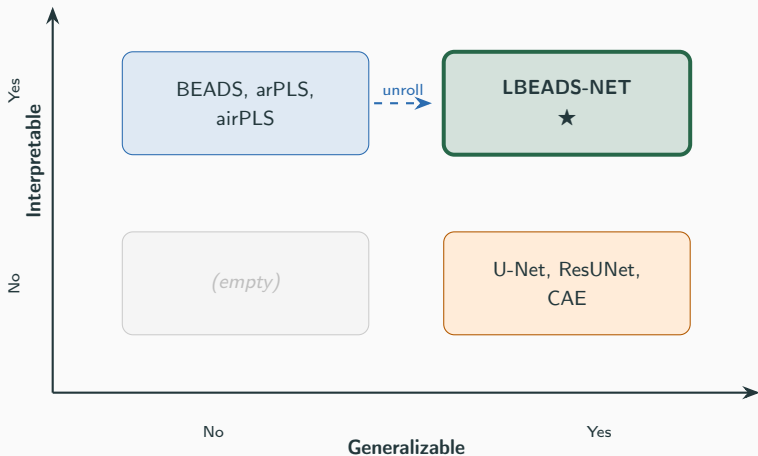
The Deep Learning Alternative

- **Convolutional autoencoders** (Kensert et al., 2021) — 190K synthetic chromatograms
- **ResUNet** (Chen et al., 2022) — 2.35M parameters, 210K spectra
- **DIRAS+** (Aradhye et al., 2025) — hybrid CNN + classical solver

These methods generalize without per-signal tuning...

...but they are **black boxes** with millions of opaque parameters.

The Gap

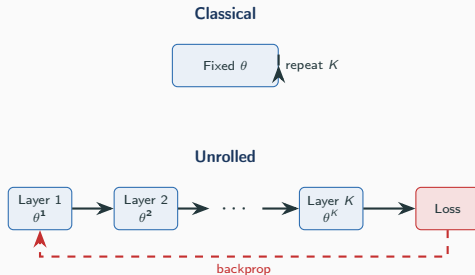


Our Approach: Algorithm Unrolling

- Map each **iteration** of an optimization algorithm to one **neural network layer**
- Algorithm parameters $\rightarrow$ **learnable** network parameters
- K layers = K iterations, trained **end-to-end**

LBEADS-NET:

20 layers $\times$ 5 parameters = **100**
learnable scalars



Act II: The Attempt

Classical BEADS (one iteration)

1. Compute reweighting matrices Λ, Γ
2. Assemble system matrix M
3. Solve banded linear system
 $(B^\top B + A^\top M A)z = d$
4. Update $x^{(k+1)} = Az$

reform.
→

ISTA Layer (one layer)

1. Gradient step:
$$x_{\text{up}} = x^{k-1} + \eta g_{\text{data}} - \eta(\nabla \mathcal{R}_1 + \nabla \mathcal{R}_2)$$
2. Proximal step:
$$x^k = \mathcal{S}_{\text{asym}}(x_{\text{up}}; \lambda_0 \eta, r)$$

Replace exact linear system solve with **gradient descent + proximal operator**.

The ISTA Layer

Gradient step:

$$\mathbf{x}_{\text{update}} = \mathbf{x}^{k-1} + \eta^k \mathbf{g}_{\text{data}} - \eta^k (\nabla_{\mathbf{x}} \mathcal{R}_1 + \nabla_{\mathbf{x}} \mathcal{R}_2)$$

Proximal step (asymmetric soft-thresholding):

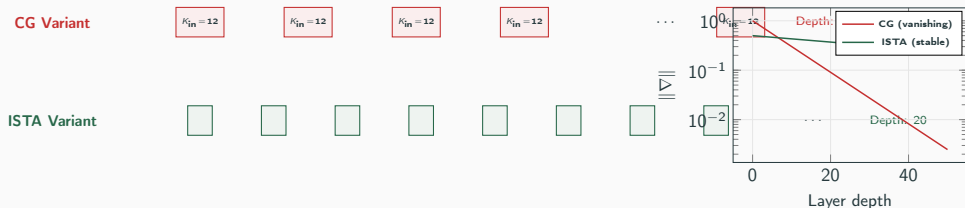
$$[\mathbf{x}^k]_i = \begin{cases} [\mathbf{x}_{\text{update}}]_i - \lambda_0^k \eta^k & \text{if } [\mathbf{x}_{\text{update}}]_i > \lambda_0^k \eta^k \\ [\mathbf{x}_{\text{update}}]_i + \lambda_0^k \eta^k r^k & \text{if } [\mathbf{x}_{\text{update}}]_i < -\lambda_0^k \eta^k r^k \\ 0 & \text{otherwise} \end{cases}$$

$\lambda_0 \eta$ — **effective threshold**: controls sparsity level

r^k — **asymmetry ratio**: suppresses negative artifacts

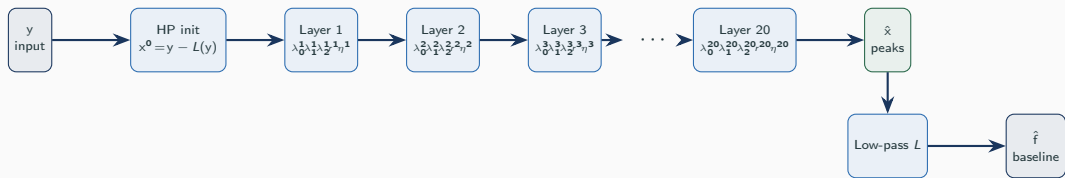
Warning: The proximal operator shrinks **ALL** non-zero values $\rightarrow$ source of baseline leakage

Why Not Conjugate Gradient?



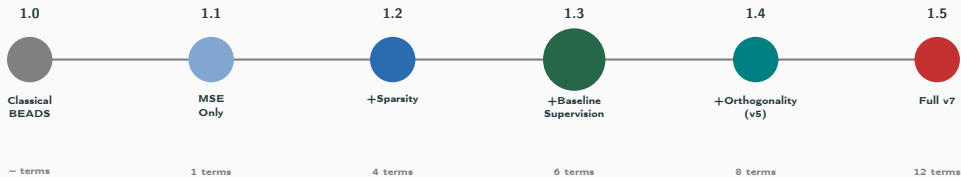
Simpler layers $\times$ more layers $>$ Complex layers $\times$ fewer layers

Full Architecture



Signal length $N = 4096$ | $K = 20$ layers | 5 params/layer | **100 total learnable scalars**

The Development Timeline



Progressive Development: Each experiment adds one key idea

Exp 1.0: Classical BEADS (Baseline)

- Apply classical BEADS with published default parameters
- Parameters designed for unnormalized data (amplitudes ~ 100 s)
- Applied to normalized $[0, 1]$ test signals

Metric	Value
Area Error	99.99%
Peak Correlation	0.746
Baseline Leakage Index	0.837

Almost all peak content treated as baseline.

Takeaway: Default parameters catastrophically fail on normalized data — **the tuning problem in its most acute form.**

Exp 1.1: Naive Unrolling (MSE Only)

Train with reconstruction loss only: $\alpha_{\text{mse}} = 1.0$, all others = 0.

Metric	Classical	Exp 1.1
Peak MSE	21.27	9.79 ✓
Peak Corr.	0.746	0.748
Area Error	99.99%	49.47% ✓
BLI	0.837	0.593 ✓
Neg. Fraction	0.000	0.295 ✗

Algorithm unrolling works — but the architecture has an inherent sparsity bias.

Exp 1.2: + Sparsity Priors

Add ℓ_1 sparsity, total variation, non-negativity penalties.

Improved:

Peak Corr: 0.748 $\rightarrow$ **0.785** ✓

Neg. Fraction: 0.295 $\rightarrow$ **0.216** ✓

Unchanged:

BLI: 0.593 $\rightarrow$ 0.597 ✗

Area Error: 49.47% $\rightarrow$ 49.51% ✗

Sparsity priors improve peak *shape* but do **NOT** address the leakage mechanism.

Exp 1.3: + Baseline Supervision

Add masked baseline reconstruction + baseline smoothness.

Metric	Exp 1.2	Exp 1.3
Peak MSE	9.79	9.30
Peak Corr.	0.785	0.793
Baseline MSE	—	7.48

The single most impactful addition.

Direct gradient signal about baseline quality provides information that pure peak reconstruction cannot.

Only 6 active loss terms.

Exp 1.4: + Orthogonality (v5 — Thesis Model)

Add peak–baseline orthogonality + baseline third-derivative penalty.

Metric	Exp 1.3	Exp 1.4
Baseline MSE	7.48	7.40 ✓
BLI	0.596	0.592 ✓
Peak Corr.	0.793	0.785 (mild decrease)

Enforcing mutual exclusivity between peaks and baseline improves baseline quality at a mild cost to peak shape.

This is the thesis model configuration.

Exp 1.5: Full Anti-Leakage Battery (v7)

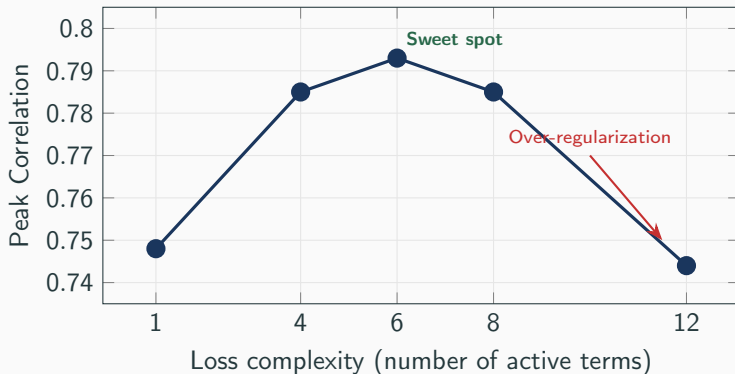
Activate **ALL 12 loss terms**: asymmetric baseline, leakage penalty, envelope constraint, frequency separation.

Metric	Exp 1.4	Exp 1.5
Peak Corr.	0.785	0.744 ✗
Peak MSE	9.36	10.01 ✗
Neg. Fraction	0.211	0.305 ✗
BLI	0.592	0.611 ✗

More loss terms $\neq$ better performance.

12 competing objectives overwhelm 100 parameters in 300 gradient steps.

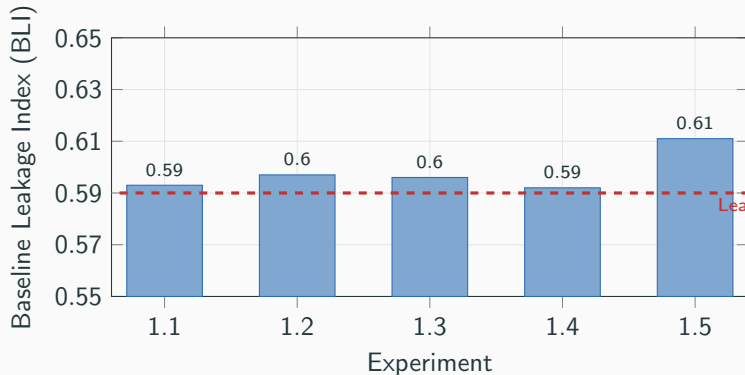
The Over-Regularization Lesson



The relationship between loss complexity and model quality is **non-monotonic**.

This motivated the 3-stage curriculum training strategy.

The BLI Plateau



BLI varies by only **0.005** across Experiments 1.1–1.4.

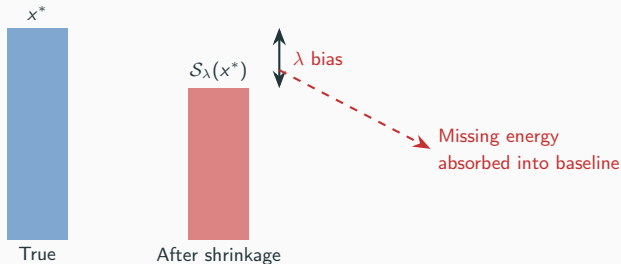
Loss engineering cannot overcome the architectural inductive bias.

Act III: The Discovery

Why Leakage is Fundamental

Soft-thresholding:

$$\mathcal{S}_\lambda(x) = \text{sign}(x) \cdot \max(|x| - \lambda, 0)$$



For **any** $\lambda > 0$, every non-zero signal value is shrunk by exactly λ .

In dense-peak regions where the sparsity assumption fails, this shrinkage accumulates across all peak samples.

Three Root Causes of Baseline Leakage

Sparsity Assumption Violation

Dense peaks overlap $\rightarrow$ signal is NOT sparse
 ℓ_1 over-shrinks $\rightarrow$ energy leaks to baseline

Fixed Cutoff Frequency

$f_c = 0.006$ for ALL signals
Broad peaks have energy below $f_c \rightarrow$ attributed to baseline

Spatially Uniform Regularization

Same λ everywhere
Dense regions need weak λ , sparse regions need strong λ

All three compound in **dense-peak regions** — where accurate baseline correction is most needed.

Emergent Parameter Specialization

Parameter	Layer 0	Layer 3	Layer 5
λ_0 (sparsity)	0.481	1.198	2.000*
r (asymmetry)	3.36	2.00*	9.67
η (step size)	0.248	0.191	0.243

- **Early layers:** gentle, conservative separation (low λ_0 , moderate r)
- **Late layers:** aggressive refinement (high λ_0 , strong asymmetry)
- **Step size:** determined by filter spectral properties, not layer position

This specialization **emerges from training** — all layers initialized identically.

What Actually Worked

- ✓ Algorithm unrolling eliminates per-signal tuning
(2× MSE improvement over default BEADS)
- ✓ Masked baseline supervision is the single most impactful loss term
- ▲ Baseline leakage persists at $\text{BLI} \approx 0.59$ — a fundamental ℓ_1 limit
- ✗ More loss terms $\neq$ better: over-regularization degrades performance
- ✓ Learned parameters exhibit emergent layer specialization

The negative results (3, 4) are **as important as the positive ones**.

Limitations

- Synthetic-only evaluation — no real chromatographic ground truth
- No comparison with arPLS, airPLS, or deep learning baselines
- Dense $O(N^2)$ matrices — 128 MB per buffer at $N=4096$
- Fixed cutoff frequency f_c (not learnable due to SciPy–PyTorch boundary)
- $\sim 50\%$ area error — not deployment-ready for quantitative analysis
- 40 test signals — limited statistical power

This is a **methodological exploration**, not a production tool.

Future Directions

1. Group sparsity ($\ell_{2,1}$ penalty)

Replace element-wise ℓ_1 with block soft-thresholding.

ℓ_1 : shrinks each element vs. Group: preserves within-group amplitudes.

Directly addresses the leakage mechanism.

2. Banded matrix implementation

$O(N^2) \rightarrow O(N \cdot b)$. Enables $N = 100,000+$ signals.

3. Real chromatographic validation

Spike-and-recovery experiments, consensus labeling.

The critical missing piece.

Thank You

Algorithm unrolling bridges interpretability and generalization —
and reveals the fundamental limits of sparsity-based signal decomposition.

Advisor: Professor Ivan Selesnick

Saad Zubairi — NYU Tandon — May 2026

Questions?